

Aqueous humour and glaucoma mechanisms

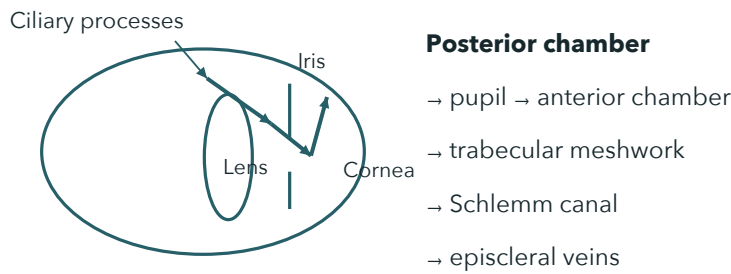
Pre-clinicals | Pressure, drainage and ganglion cell injury

Follow the fluid

Two aqueous chambers

The anterior chamber lies between cornea and iris. The posterior chamber is the narrow space behind the iris and in front of the lens; it is distinct from the vitreous cavity. Aqueous humour nourishes the avascular cornea and lens, removes metabolites and helps maintain intraocular pressure (IOP).

Production and drainage



Ciliary processes → posterior chamber → pupil → anterior chamber.
Conventional outflow: trabecular meshwork → Schlemm canal → collector channels → episcleral veins. Alternative outflow: through ciliary muscle → suprachoroidal space → sclera.

Production is active

- Non-pigmented ciliary epithelium of the pars plicata secretes aqueous; the ciliary muscle changes lens shape.
- Ion transport, including carbonic anhydrase dependent bicarbonate formation, draws water into the posterior chamber.
- Trabecular outflow is the major route. Uveoscleral outflow is the alternative route; their exact proportions vary with age and other factors.

What raises pressure

Change	Effect on IOP
Higher aqueous inflow	Tends to increase IOP
Higher trabecular resistance	Reduces drainage; IOP rises
Higher episcleral venous pressure	Reduces the drainage pressure gradient
Greater outflow facility	Tends to reduce IOP

Pressure is a risk factor

Glaucoma is optic nerve damage

Glaucoma causes progressive retinal ganglion cell and axon loss, with characteristic optic disc and retinal nerve fibre layer changes. IOP is an important modifiable risk factor, but pressure alone cannot establish or exclude the disease. Assess structure, function and the drainage angle together.

Keep the categories separate

Pattern	Interpretation
High IOP; no detectable glaucomatous injury	Ocular hypertension
Open angle; glaucomatous injury; raised IOP	Open angle glaucoma
Open angle; glaucomatous injury; IOP consistently in the usual range	Normal tension glaucoma
Large cup; no other convincing injury	May be physiological; disc size and serial change matter

Structure predicts function

- Ganglion cell axons pass through the lamina cribrosa
- ↓
- Mechanical and vascular stresses contribute to axonal injury
- ↓
- Neuroretinal rim thins or develops focal notching
- ↓
- Nerve fibre loss produces corresponding visual field defects

High yield pairing

- Inferior retinal nerve fibre loss produces a superior visual field defect; superior loss produces an inferior defect.
- Arcuate scotomas and nasal steps follow retinal nerve fibre bundle anatomy and often respect the horizontal meridian.
- Central acuity can remain good despite substantial field loss. A reading chart alone misses this damage.
- Disc asymmetry, rim notching and progression are more informative than a single cup-to-disc cut-off.

Measurement trap

IOP varies over time, and corneal thickness and biomechanics affect tonometry. A single normal reading does not rule out glaucoma.

Open angle or blocked angle

Locate the obstruction

Open angle mechanism	Angle closure mechanism
Trabecular meshwork remains exposed	Peripheral iris covers trabecular meshwork
Increased functional outflow resistance	Physical iridotrabecular contact
Often gradual and initially silent	Can be acute, intermittent or chronic
Ganglion cell injury can occur at normal IOP	Pressure can rise sharply when drainage is obstructed

Relative pupillary block

- Aqueous passage between lens and iris is impeded
- ↓
- Posterior chamber pressure exceeds anterior chamber pressure
- ↓
- Peripheral iris bows forward: iris bombe
- ↓
- Iris contacts the trabecular meshwork
- ↓
- Outflow falls and IOP rises

Recognise the anatomical setup

- Short hypermetropic eyes, shallow anterior chambers and thicker ageing lenses favour a narrow angle.
- Mid-dilation can increase iris-lens contact and crowd an already narrow angle.
- Acute marked IOP elevation can cause corneal oedema, blurred vision, coloured halos, ocular pain and nausea.
- Angle closure does not always involve pupillary block. Plateau iris and lens related crowding are alternative mechanisms.

Why an iris opening helps

Peripheral iridotomy bypasses pupillary block by connecting the two aqueous chambers and equalising pressure. It does not repair trabecular dysfunction or established peripheral anterior synechiae.

Classification trap

An acute pressure rise is not itself proof of established glaucomatous optic neuropathy. Angle closure and angle closure glaucoma are distinguished by whether glaucomatous injury is present.

Mechanisms you can apply

Drug mechanism, not a treatment sequence

Class / example	Pressure lowering mechanism
Beta blocker: timolol	Reduces aqueous production by ciliary epithelium
Carbonic anhydrase inhibitor: dorzolamide, acetazolamide	Reduces bicarbonate dependent aqueous secretion
Alpha-2 agonist: brimonidine	Reduces secretion; also increases uveoscleral outflow
Prostaglandin analogue: latanoprost	Mainly increases uveoscleral outflow
Muscarinic agonist: pilocarpine	Contracts ciliary muscle, improving trabecular outflow; also causes miosis

Secondary pressure elevation

Cause	Mechanism
Corticosteroid exposure	Changes trabecular extracellular matrix and cell function; outflow resistance rises
Retinal ischaemia with iris new vessels	Fibrovascular tissue impairs drainage and may contract to close the angle
Anterior uveitis	Cells/debris obstruct outflow; synechiae can cause pupillary block
Elevated episcleral venous pressure	Higher downstream pressure reduces conventional outflow

Mechanism checks

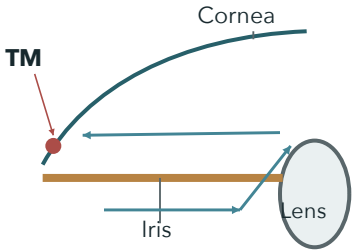
- Timolol is non-selective: systemic absorption can cause bronchospasm or bradycardia.
- An inflamed ciliary body may initially produce less aqueous, so uveitis can lower IOP despite its potential to cause secondary pressure elevation.
- Loss of ganglion cells is irreversible. Lowering IOP aims to reduce further injury, not restore axons already lost.

Visual mechanisms and references

Original educational schematics. Colours and proportions are simplified.

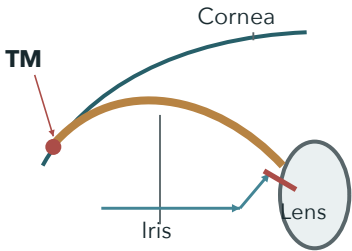
Compare the drainage angle

OPEN ANGLE



Iris stays clear of the TM.
Drainage resistance can still be high.

ANGLE CLOSURE

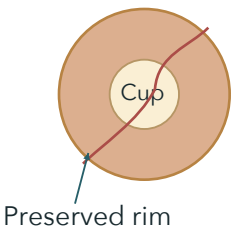


Peripheral iris covers the TM.
Pupillary block can bow iris forward.

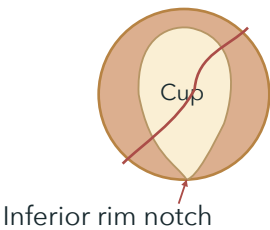
TM = trabecular meshwork. Blue arrows show aqueous movement. The open angle can still drain poorly; angle closure adds physical iris obstruction. [G1]

Read the rim, not just the cup

NORMAL ANATOMY



GLAUCOMATOUS PATTERN



Neuroretinal rim = tissue between cup and disc edge

Inferior rim loss predicts a superior field defect.

These disc sketches show anatomy and a pattern of rim loss. Cup size alone is insufficient; disc size, progression and matching fields matter. [G2]

References for these figures

- G1. National Eye Institute, Types of glaucoma: angle and pressure mechanisms.
- G2. AAO Glaucoma Committee, Primary Open-Angle Glaucoma PPP (2026): disc, rim and nerve fibre assessment.